STANDARDLIFTERS

MAKING INNOVATION THE STANDARD®

Why Use **Stock Ejectors**

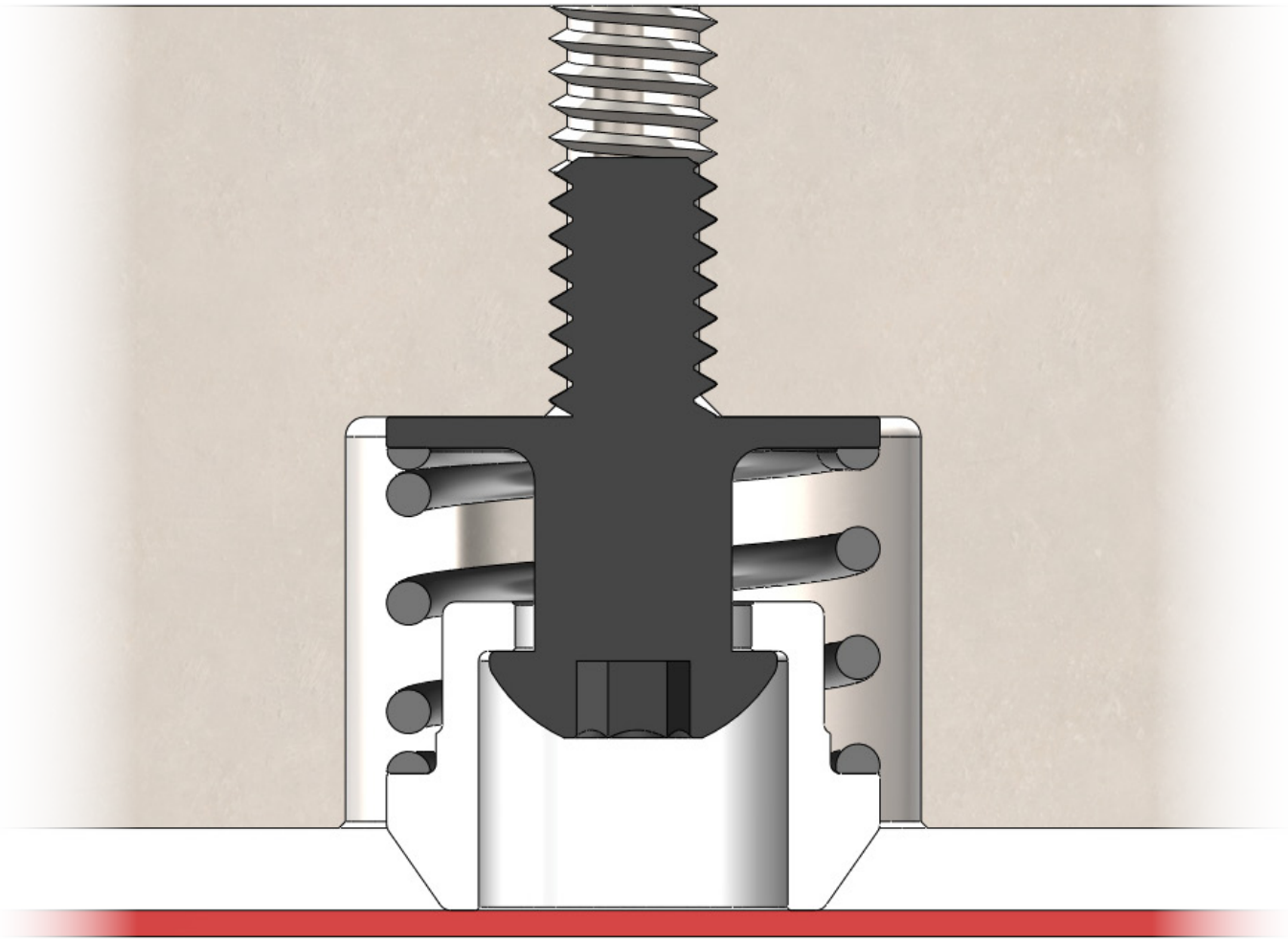


The New Industry Standard

This product may be covered by one or more patents or patent applications.
See http://www.standardlifters.com/patents.html for details.

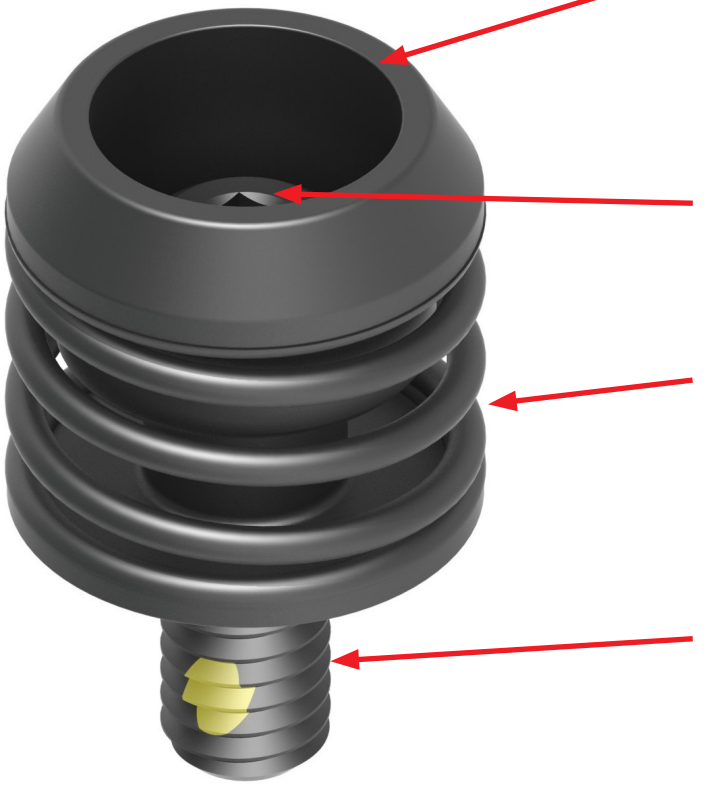
8/14 - Rev. 1

Application - **Ejecting Sheet Metal**



Prevents lubricated sheetmetal stock from sticking to large surfaces in tooling, such as Stripper Pads and Form Pads.

Product Features / Value



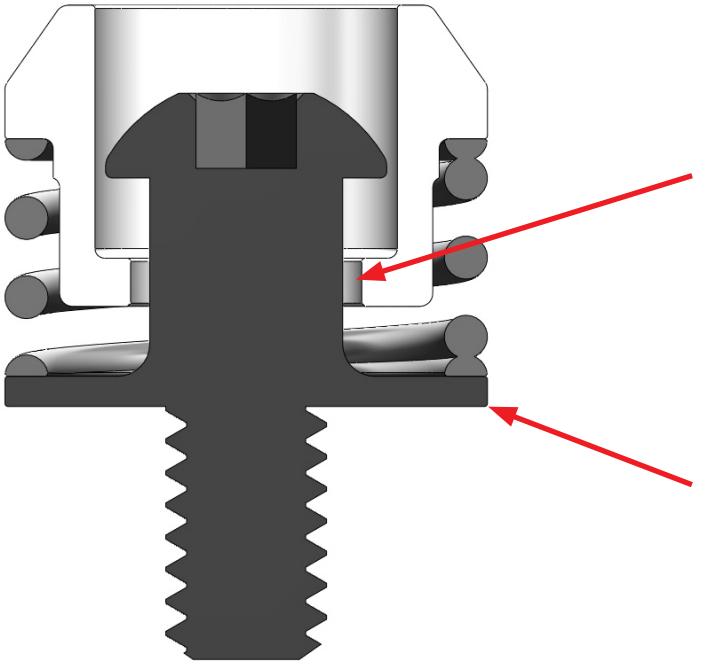
Large Stripping Surface
to distribute force which provides optimal performance, and lower potential for marking the part.

Easy Assembly with Standard Tools
using a standard Hex wrench to install

Large Spring
provides long life performance. Designed for and tested to, millions of strokes

Anti-Loosen Patch
ensures the assembly stays in place, but is still accessible from the working side of the tool

No Alterations Needed
such as grinding Ejector Pins to length, or cutting springs



Tippable / Flexible
clearances in the assembly allow the stripper to tip. This helps prevent damage to the assembly, even if it is bumped during the construction process

Complete Asssembly - No Loose Parts
no loose springs or set screws that can be easily lost when assembling

Method Comparison



- Large Spring - long life

- No Loose Parts - ready to bolt down

- Large Stripping Surface - less likely to mark part

- Standard Tools to Install - hex wrench

- No Alterations Required

- Assembly clearances allow the stripper to tip without damage if it gets bumped

- Assembly is less prone to damage if debri gets into the assembly due to clearances



- Small Spring - known to fail - low life

- Small Point Contact can mark the part

- Some styles require “special” tools to install

- Tip can be bent if it gets bumped by another detail during the construction process

- Debri in the assembly can cause premature failure due to the tighter clearance between the body and the point



- Small Spring - known to fail - low life

- Loose Parts - harder to handle and easier to lose

- Higher Labor Cost - likely requires ejector pins to be altered (cut to length and debur)

- Ejector can be bent if it gets bumped by another detail during the construction process

- Takes more time (higher cost) to assemble and disassemble because it is installed from the “backside” of the detail

Force vs. Contact Surface Area

The SL Stock Ejector is designed with adequate force to ensure consistent stock ejection every stroke / hit. Due to the large contact surface area, the SL Stock Ejector is less likely to mark the stock / part.

	Stock Ejector		Spring/Nitro Plunger			
Max End Force	Contact Surface Area	Resulting LBS per Sq. Inch	Contact Surface Area	Resulting LBS per Sq. Inch	Difference	Force to be comparable to Stock Ejector
12.5 lbs. (.75" Dia. Stock Ejector Comparison)	 .060 square inches	**208** LBS per Square Inch	 .100" Diameter .008 square inches	**1602** LBS per Square Inch	**7.7x** times higher	**1.62 lbs.**
18.7 lbs. (.875" Dia. Stock Ejector Comparison)	 .088 square inches	**212** LBS per Square Inch	 .100" Diameter .008 square inches	**2397** LBS per Square Inch	**11.3x** times higher	**1.66 lbs.**
19.6 lbs. (1.0" Dia. Stock Ejector Comparison)	 .128 square inches	**153** LBS per Square Inch	 .100" Diameter .008 square inches	**2513** LBS per Square Inch	**16.4x** times higher	**1.19 lbs.**

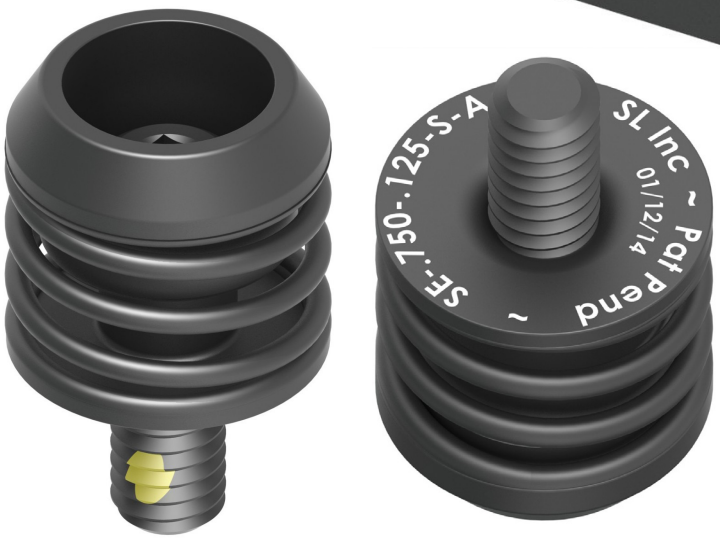
Example: If you had a 1" square block in your hand that weighed 1 pound, you would feel the pressure of 1 pound per square inch. But, if you had a 2" square block in your hand that weighed 1 pound, you would feel the pressure of 1/4 pound per square inch because the area has quadrupled. It is the same amount of pressure, just spread over a larger area.



Product Offering / Pricing

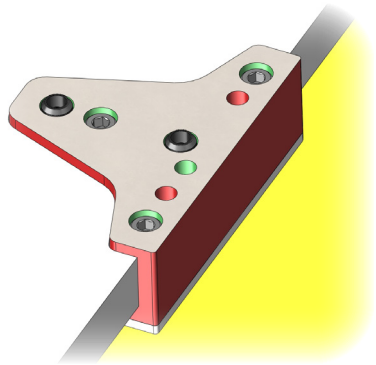
Overall Diameter	Thread Size	List Price
.750 / 20mm	1/4-20 / M6-1.0	$7.95
.875 / 22.5mm	5/16-18 / M8-1.25	$8.95
1.000 / 25mm	3/8-16 / M10-1.5	$9.95



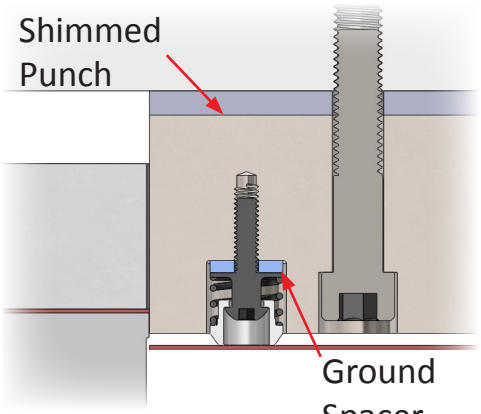


Additional Option - Punch Slug Ejector

Application - Punches that require sharpening



This style is designed to be used on surfaces that will be ground for sharpening purposes. It allows for the Stock Ejector assembly to be "shortened" when the punch is sharpened, so the travel and force is consistant throughout the die life of the punch.



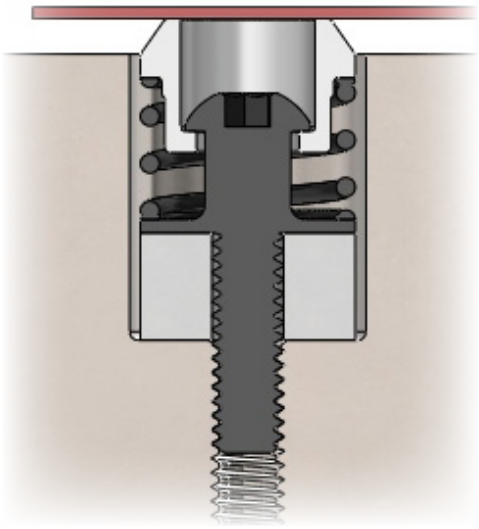
Product Features / Function



Provides all the same features and benefits as the "standard" stock ejector.

- Large Spring - Ensures long life
- Complete Assembly - no loose springs or set screws
- No Alterations Needed - No grinding ejectors to Length

Assembly comes with a "spacer" that is shortened at the same time the punch is sharpened. By grinding the spacer the same amount as the punch, the assembly maintains the same relationship to the punch surface. This provides consistant and reliable operation for the life of the punch. The spacer's starting thickness is 3/8" or 10mm which accommodates for typical industry "die life" on punches.

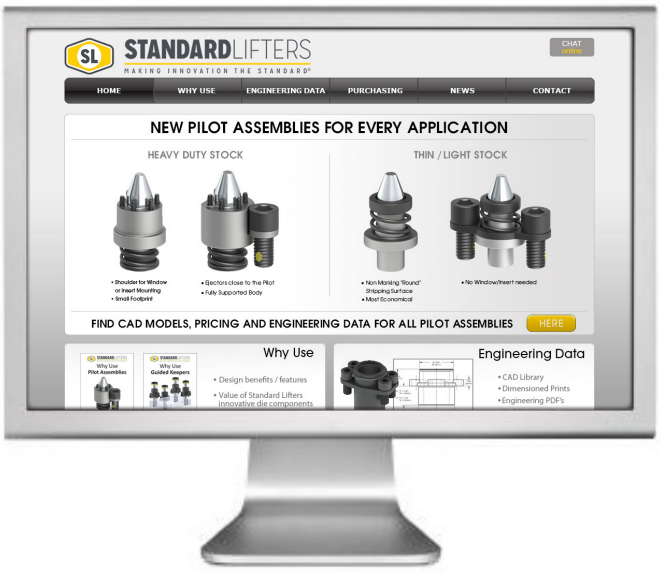


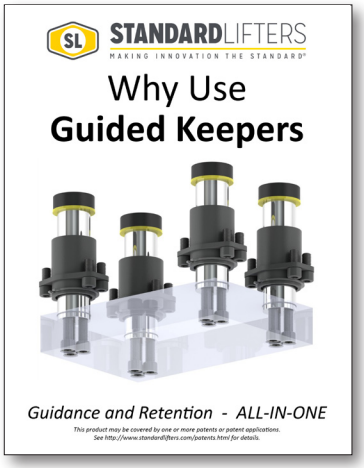
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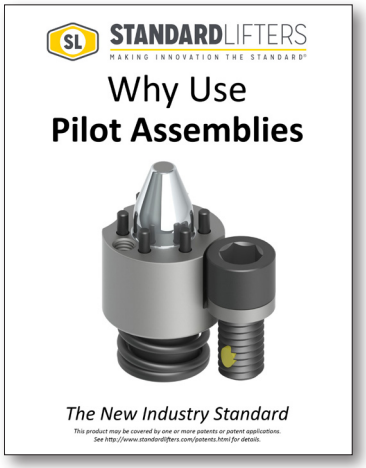
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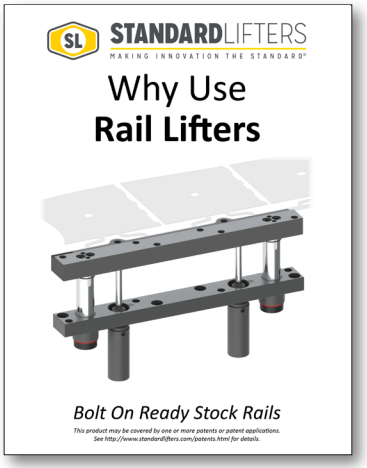
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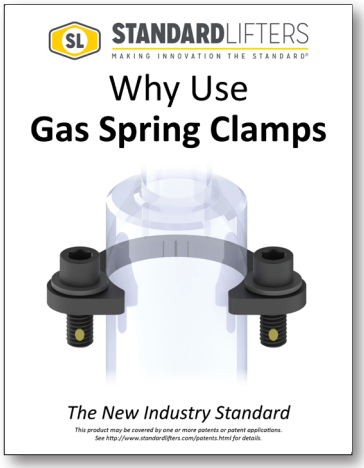
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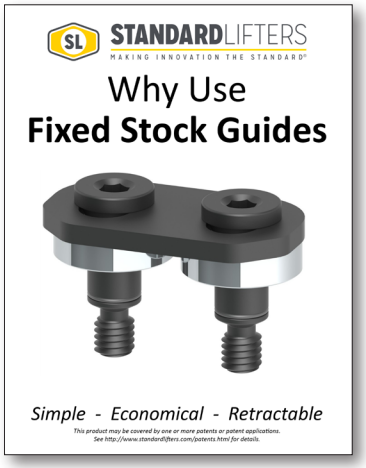


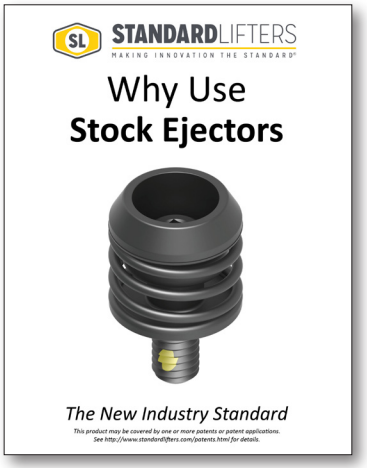








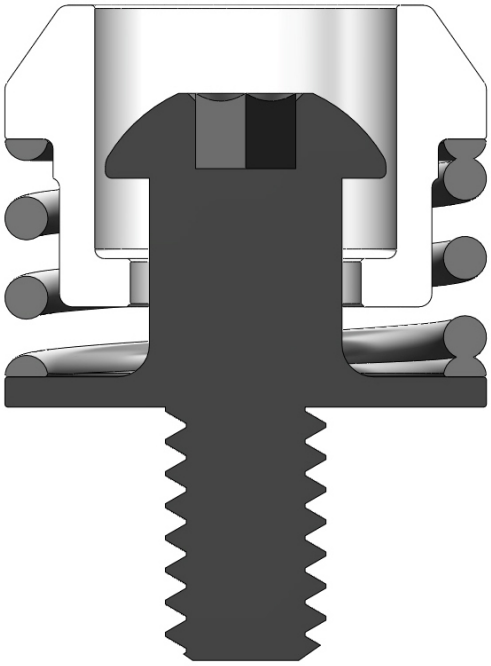


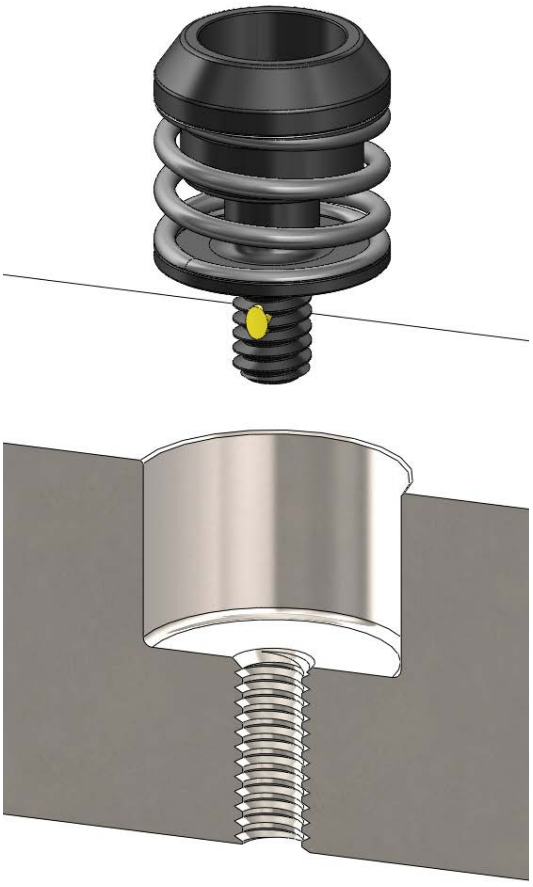


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MAKING INNOVATION THE STANDARD®

Engineering Data
Stock Ejectors





The New Industry Standard

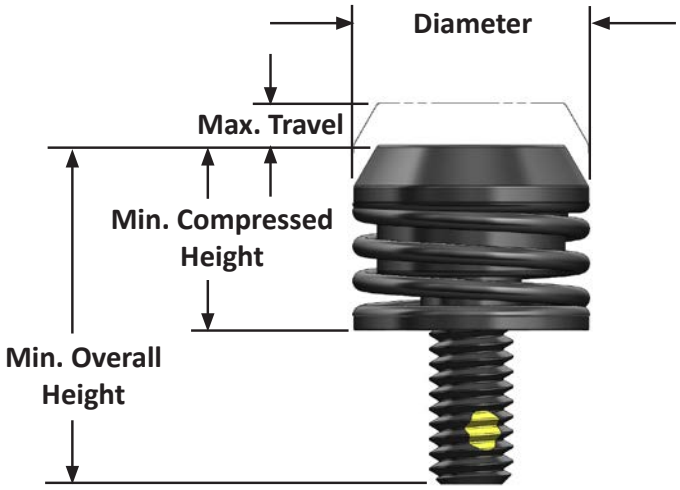
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Things to Consider

When choosing the right Stock Ejector for your design.

Space Constraints - Assembly Diameter & Height



Outside Diameter	Minimum Compressed Height	Minimum Overall Height
.750”	.625”	1.02”
.875”	.688”	1.26”
1.00”	.750”	1.25”
20mm	15.88mm	24.8mm
22.5mm	17.48mm	31.6mm
25mm	19.05mm	31.4mm

Fastener Size - Travel - Stripping Force

Outside Diameter	Thread Size	Max. Travel	Force Range
.750”	1/4-20	.125”	6-12 lbs
.875”	5/16-18	.156”	13-19 lbs
1.00”	3/8-16	.188”	14-20 lbs
20mm	M6-1.0	3.2mm	6-12 lbs
22.5mm	M8-1.25	4mm	13-19 lbs
25mm	M10-1.5	4.8mm	14-20 lbs

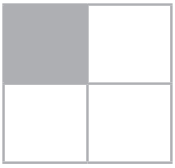


Force vs. Contact Surface Area

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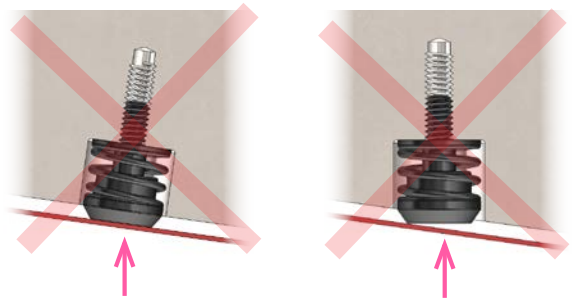
Example: If you had a 1" square block in your hand that weighed 1 pound, you would feel the pressure of 1 pound per square inch. But, if you had a 2" square block in your hand that weighed 1 pound, you would feel the pressure of 1/4 pound per square inch because the area has quadrupled. It is the same amount of pressure, just spread over a larger area.



Recommendations

Use of SE Assemblies on Angled Surfaces is NOT recommended

Standard Lifters does not recommend using Stock Ejector units on angled surfaces or where the material is compressing the stripper on an angle.
Doing so may cause damage to the Stock Ejector assembly and may fail to function properly.



Recommended Torque Values

SE Unit	Thread Size	Max. Torque (in.- lbs.)
.750	1/4-20	125
.875	5/16-18	200
1.000	3/8-16	360

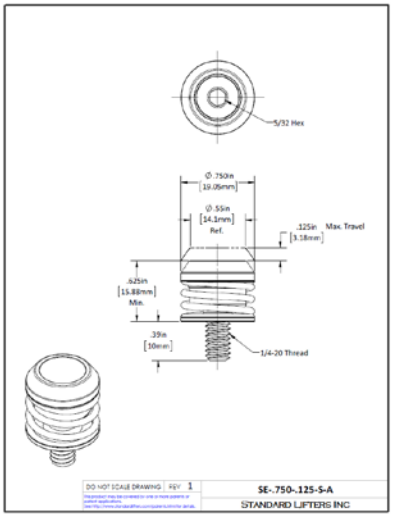
SE Unit	Thread Size	Max. Torque (in.- lbs.)
20mm	M6-1.0	88
22.5mm	M8-1.25	212
25mm	M10-1.5	425

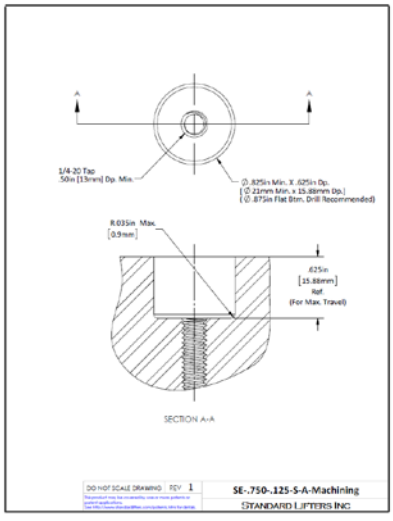
Recommended Replacement Interval

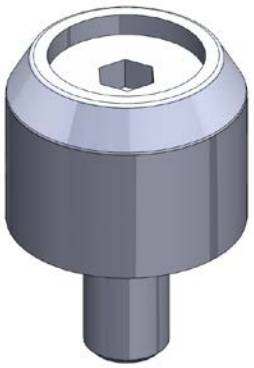
To ensure the function of stock ejection in the die application is maintained, Standard Lifters recommends replacing the Stock Ejector unit every **2 million** strokes. It is better to replace it while it is in good working order than to wait until the product is not functioning as intended.

CAD Models and Dimensioned Prints

CAD Models and Dimensioned Prints are available online. Go to www.standardlifters.com







Machining Process

1. Drill Pocket



2. Spot Drill / Chamfer



3. Tap Drill



4. Tap



5. Chamfer / Debur



Stock Ejector Installed



Machine Time - Example of SE-.75

Operation	Tool Type	Speed	Feed	"In Cut" Machine Time	Tool Change Time
Drill Flat Bottom Pocket	Dual Insert Flat Bottom Drill (K-Tool FB-875)	2600RPM (600SFM)	5 IPM (.002 IPR)	9 Sec	15 sec
Spot Drill and Chamfer	Solid Carbide Spot Drill (Guhring 90072412700)	3000RPM (400SFM)	18 IPM (.003 IPR)	2 Sec	15 sec
Drill for Tap	Solid Carbide Drill (Sumitomo MDS2010MHV)	6650RPM (350SFM)	106 IPM (.008 IPR)	2 Sec	15 sec
Tap Hole	Solid Carbide Tap (Vega 84613)	500RPM (30SFM)	25 IPM (.050 IPR)	3 Sec	15 sec
Chamfer Pocket (plunge)	Dual Insert Chamfer Tool (K-Tool SPEB-2522CB)	3000RPM (800SFM)	30 IPM (.005 IPR)	2 Sec	
			Totals	**18 seconds**	**60 seconds**

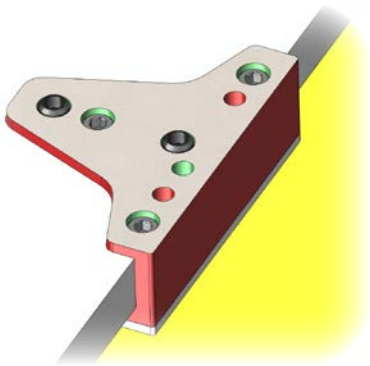
Total "In Cut" Time to Machine 1 Location = **18 Seconds**

Total Time to Machine 3 Locations in one Setup = **1 Minute and 54 Seconds**

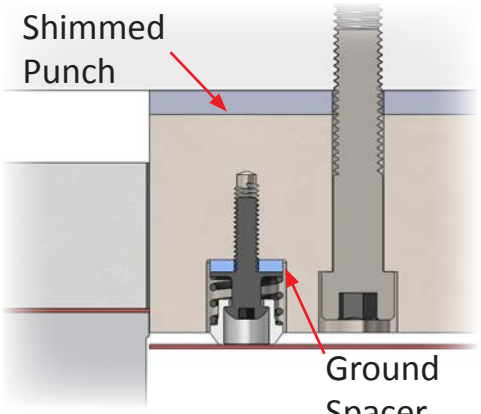
*The more locations that are machined in the same setup the lower the total machine time will be due to the reduction in time factored for tool changes.

Additional Option - Punch Slug Ejector

Application - Punches that require sharpening



This style is designed to be used on surfaces that will be ground for sharpening purposes. It allows for the Stock Ejector assembly to be "shortened" when the punch is sharpened, so the travel and force is consistant throughout the die life of the punch.

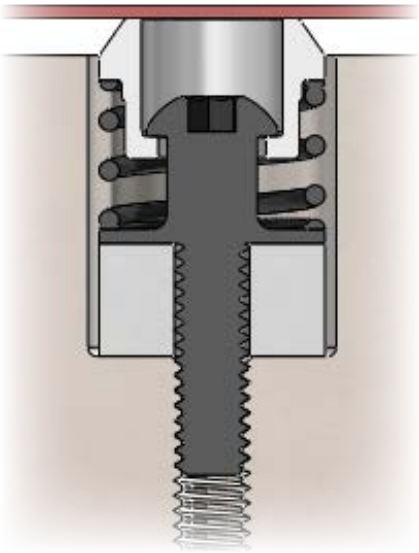


Product Features / Function



Provides all the same features and benefits as the "standard" stock ejector as well as additional machining savings.

- **Large Spring**
  - Ensures long life
- **Complete Assembly**
  - No loose springs or set screws
- **No Alterations Needed**
  - No grinding ejectors to length
- **Substantial Machining Savings**
  - No drilling deep / small diameter holes



The assembly comes with a "spacer" that is shortened at the same time the punch is sharpened. By grinding the spacer the same amount as the punch, the assembly maintains the same relationship to the punch surface. This provides consistant and reliable operation for the life of the punch. The spacer's starting thickness is 3/8" or 10mm which accommodates for typical industry "die life" on punches.

Product Offering / Pricing

Overall Diameter	Thread Size	List Price
.750 / 20mm	1/4-20 / M6-1.0	$9.95
.875 / 22.5mm	5/16-18 / M8-1.25	$11.45
1.000 / 25mm	3/8-16 / M10-1.5	$12.95

Product Number Options / Pricing

Standard Stock Ejector



Product	Overall Diameter	Max Travel	Ejection Force (Standard)	Style (A)	List Price
SE	.750	.125	S	A	$7.95
SE	.875	.156	S	A	$8.95
SE	1.000	.188	S	A	$9.95
SE	20mm	3.2mm	S	A	$7.95
SE	22.5mm	4mm	S	A	$8.95
SE	25mm	4.8mm	S	A	$9.95
Example: **SE-.750-.125-S-A**					

Punch Slug Ejector Style



Product	Overall Diameter	Max Travel	Ejection Force (Standard)	Style (A)	Option (T)	List Price
SE	.750	.125	S	A	T	$9.95
SE	.875	.156	S	A	T	$11.45
SE	1.000	.188	S	A	T	$12.95
SE	20mm	3.2mm	S	A	T	$9.95
SE	22.5mm	4mm	S	A	T	$11.45
SE	25mm	4.8mm	S	A	T	$12.95
Example: **SE-.750-.125-S-A-T**						

Packaging Structure

You will receive the product in one, or a combination of, the below packages. The packaging structure is based on the total number ordered and the best method to ship the quantity ordered in the least amount of individual packages. For example if you order 27 parts you will recieve 2 - 10packs, 1 - 5pack, and 2 - 1packs for a total of 27 individual parts.

1 Pack



5 Pack



10 Pack



For New and Updated Information

www.standardlifters.com



